Computer Network and Communication CNS3200

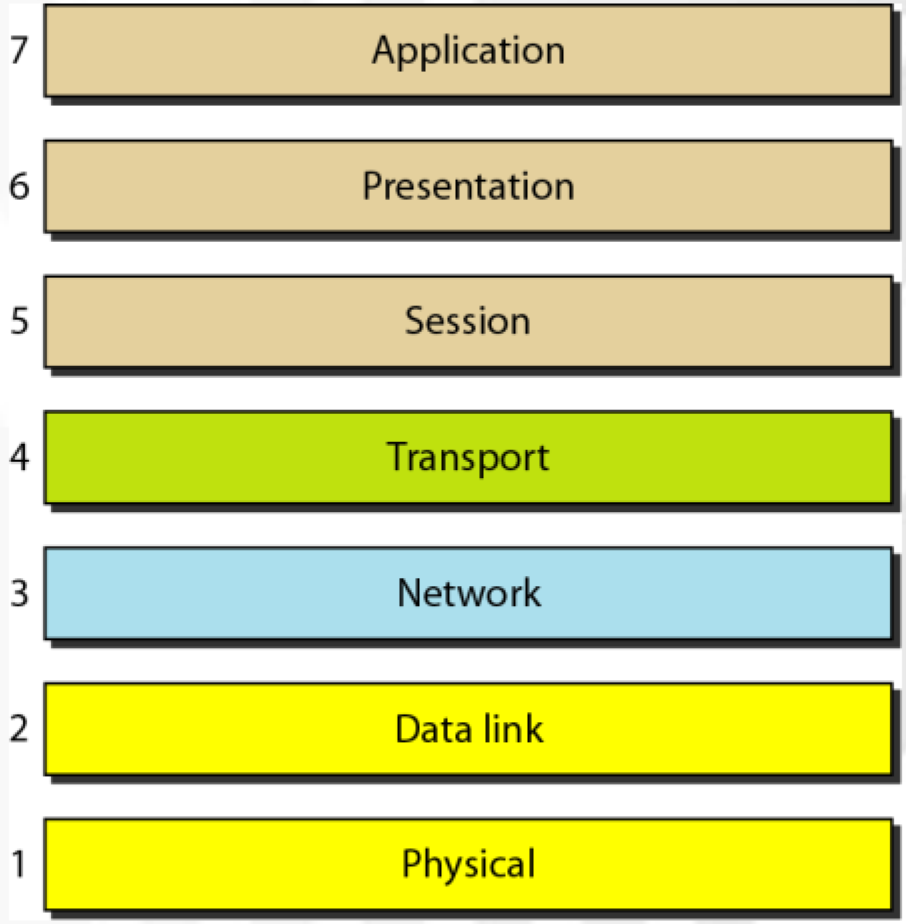
Learning Outcome

- Illustrate the layers involved in OSI model (C4)
- Illustrate the layers involved in the Internet (TCP/IP) model (C4)

The OSI Model

- Established in 1947, the ISO is a multinational body dedicated to worldwide agreement on international standard
- ISO standard which covers all aspects of network communications – **Open Systems Interconnection (OSI)** model
- **Open System** – a model that allows any two different systems to communicate regardless of their underlying architecture
- OSI is not a protocol
- Model for understanding and designing a network architecture that is flexible, robust, and interoperable

Figure 2.2 *Seven layers of the OSI model*



The model

- OSI – a layered framework for design of network systems that allows for communication across all types of computer system
- 7 layer architecture
- Intermediate nodes – involve only the first three layers
- Peer-to-Peer Process?
  - Layer n, use the services provided by layer n-1 and provides services for layer n+1
  - The process on each machine that communicate at a given layer are called peer-to-peer process
  - Use protocol

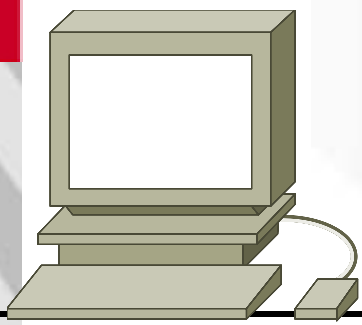
The model

- **At sending machine:**
  - headers added to the message at L6,5,4,3,2
  - Trailer is added at only L2
- **At receiving machine**
  - The message is unwrapped layer by layer, received process and removing the data meant for it.(reverse of sending machine)
- **Interfaces between layers – control the passing of the data and network information down/up through the layers of sending/receiving machine**

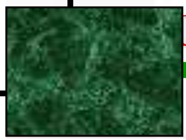
The model

- The 7L can grouped into three subgroup
  - L1,2,3 – the network support layer – deal with the physical aspects of moving data from one device to another
  - L5,6,7 – the user support layer –allow interoperability among unrelated software systems
  - L4 – ensures end-to-end reliable data transmission
- After pass through L1 of sending machine, the data unit is changed into electromagnetic signal and transported along a physical link

OSI REFERENCE MODEL

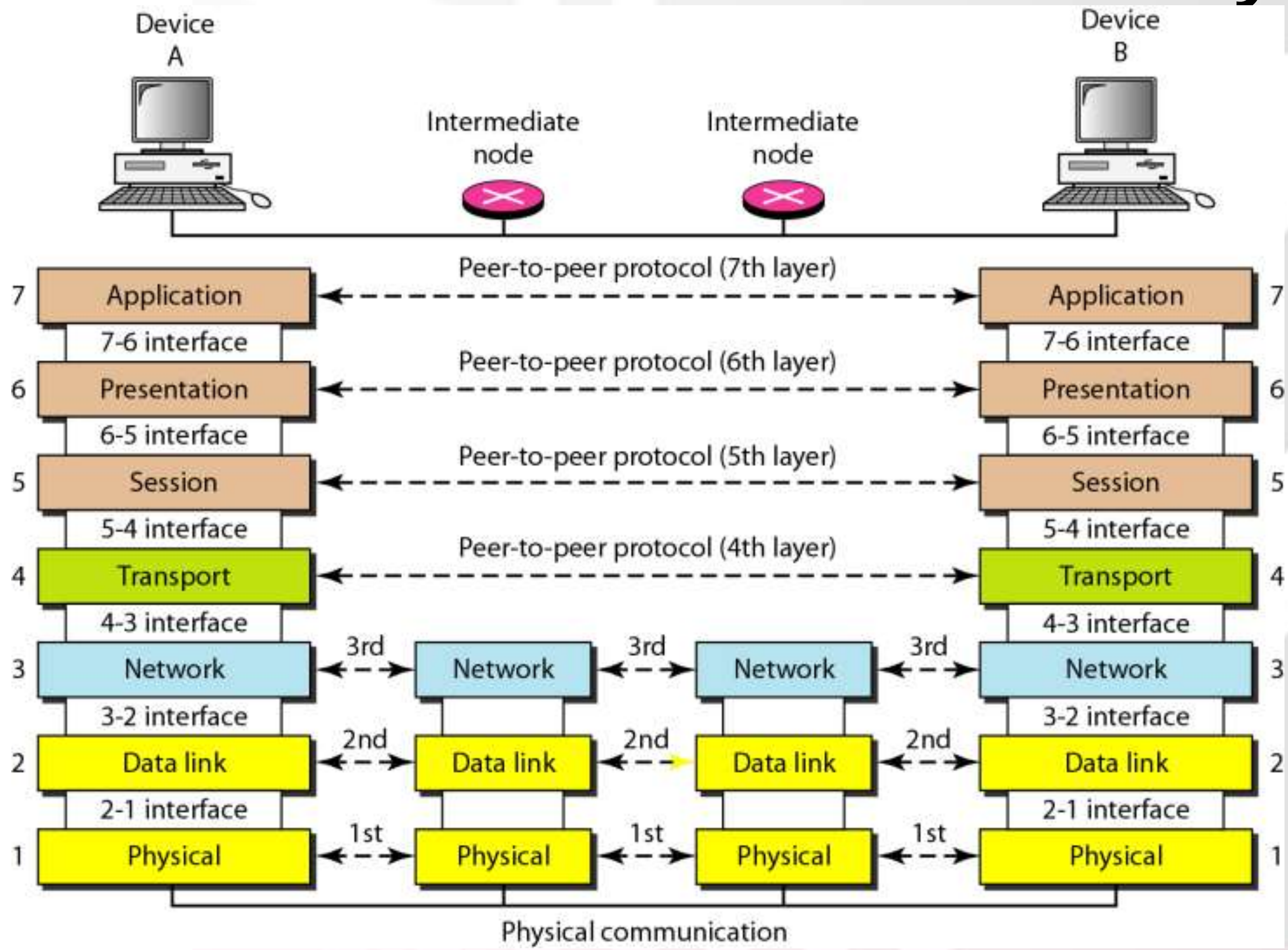


Application	Aplikasi Web, SMTP, FTP,Telnet,IRC,News dll
Presentation	Translation, Encryption and Compression
Session	Komunikasi yang selamat: Secure Socket Layer (SSL)
Transport	*Connection/less, multiplex,flow* dan *error control*
Network	Penghantaran paket ke destinasi
Data Link	Topologi Rangkaian,Pengalamatan, *Error, flow control*
Physical	DI,ATM, Ethernet, Token Ring, Dialup,wireless

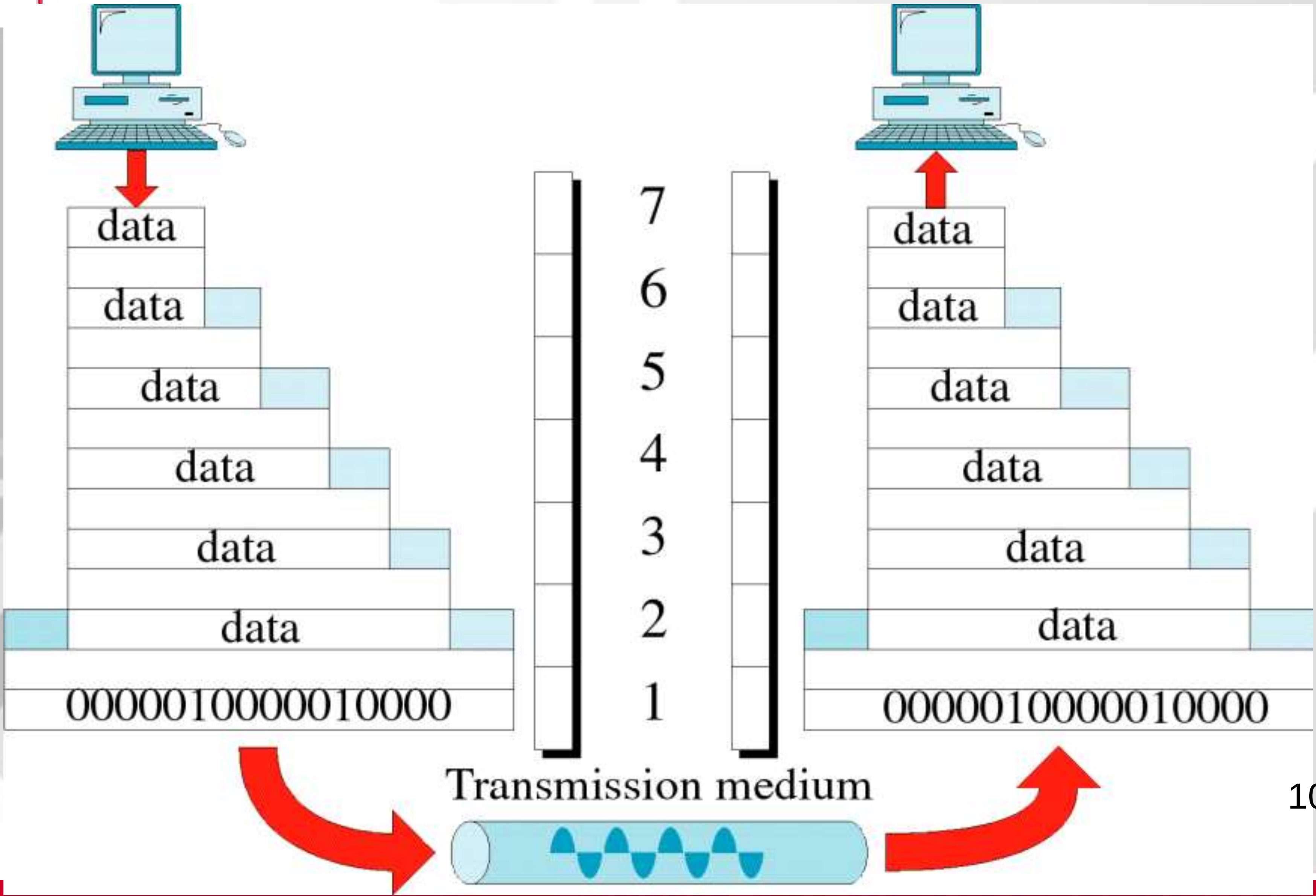


Network card

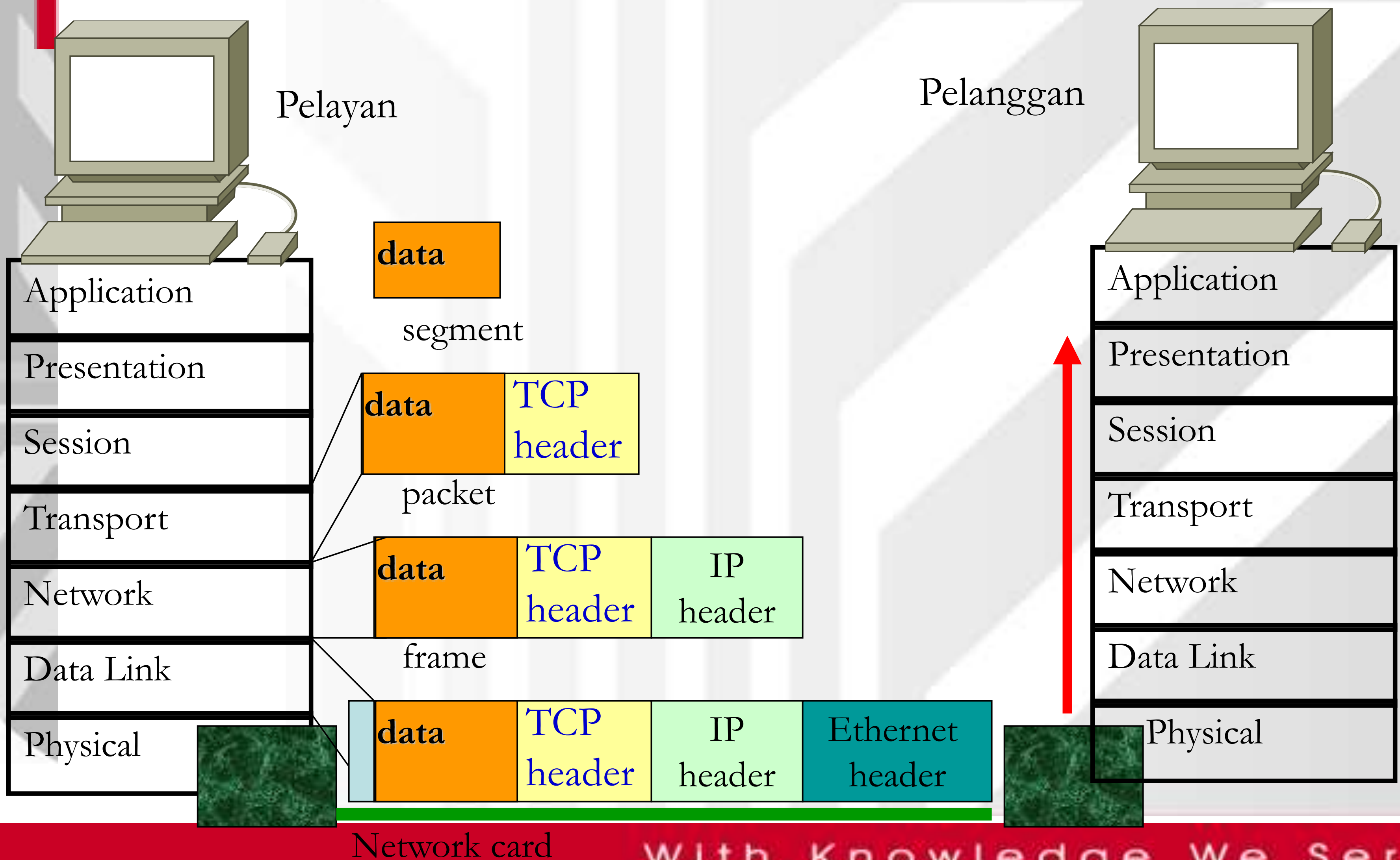
The interaction Between Layers



An Exchange Using the OSI Model



Function of *Header* in OSI and Internet



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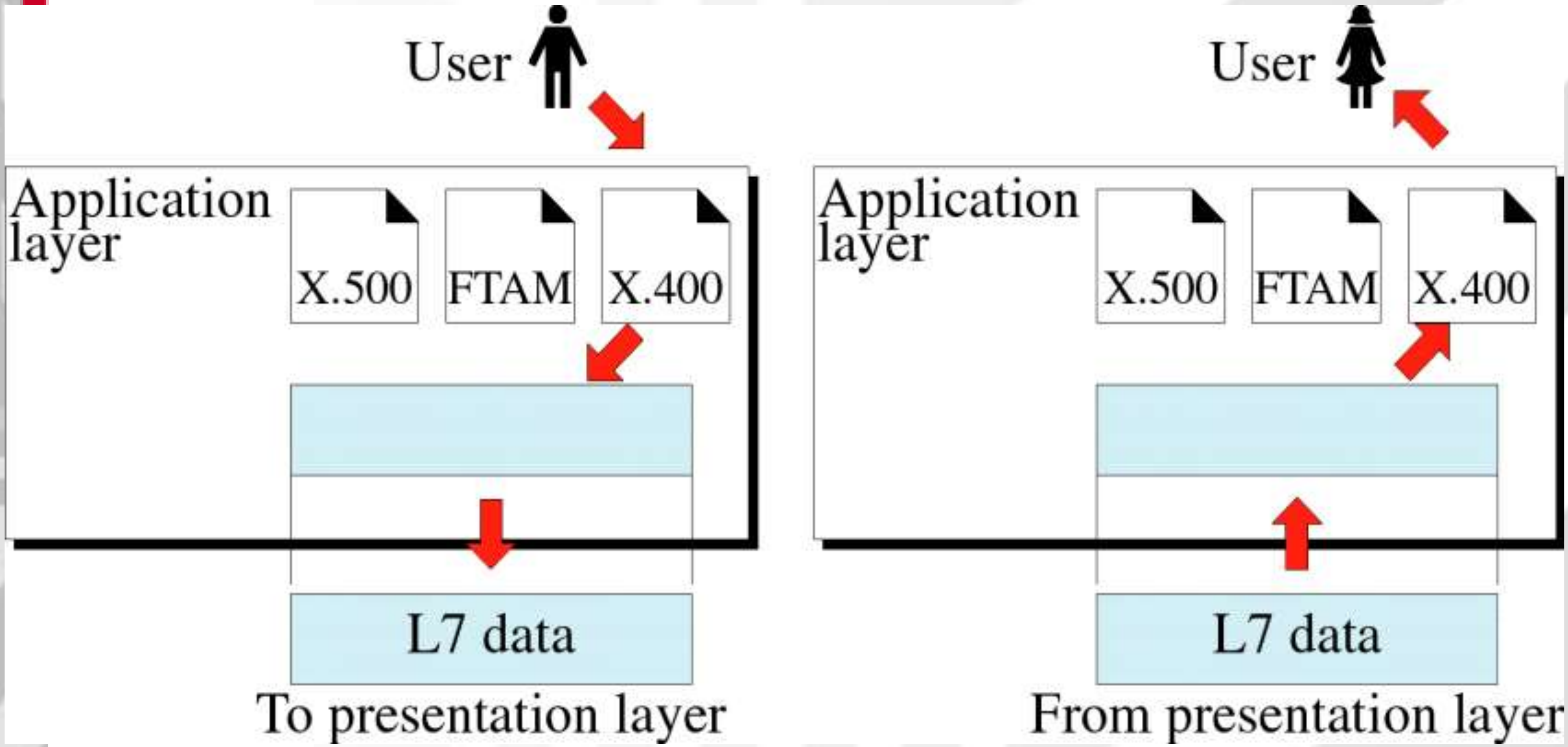
3.2 Application Layer

- Application Layer (L7)
  - Enables user , whether human or software to access the network
  - Provides user interfaces and support for services such as email, remote file access, shared database mgmt and transfer etc
  - No trailer or header are added here
- Specific services
  - Network virtual terminal
  - File transfer, access, and management (FTAM) – access/manage/control files in a remote computer
  - Mail services - X.400 – store and forward email
  - Directory services – X.500 – provides distributed database sources

Application Layer



Note:

The application layer is responsible for providing services to the user.

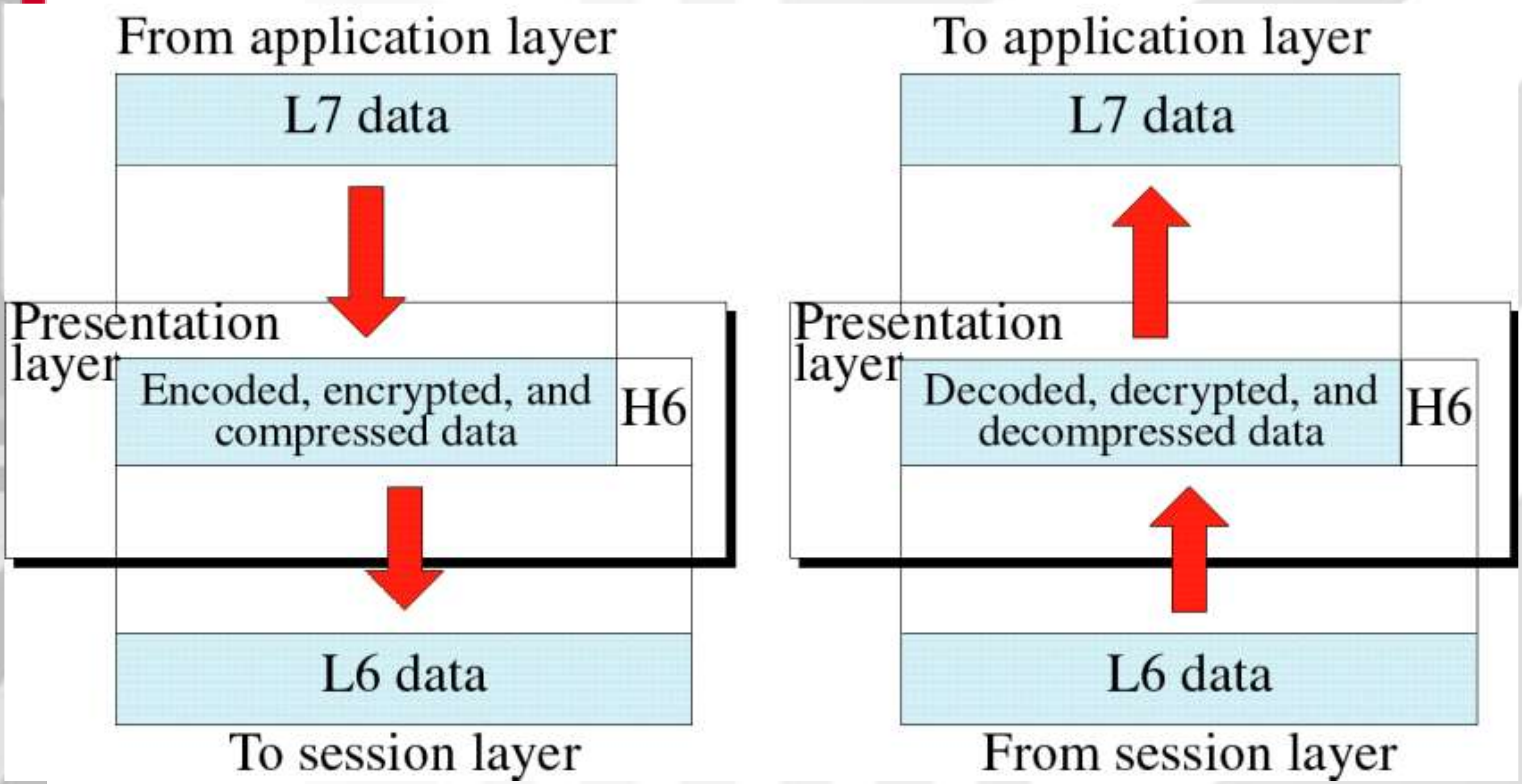
3.2 Presentation Layer

- **Presentation Layer (L6)**
  - Concerned with the syntax and semantics of the information exchanged between two systems.
- **Responsibilities:**
  - Translation
    - The process (running programs) in two systems are usually exchanging information
    - Different computers use different encoding systems
    - Responsible for interoperability between different encoding methods
    - Sender machine change the information from its sender-dependent format into a common format
    - Receiver machine change the common format into its receiver-dependent format

3.2 Presentation Layer

- Encryption
  - Encryption - transform the original information to another form and sends it over the network
  - Decryption - reverse process at the receiver side
  - assure privacy - to carry a sensitive data/information
- Compression
  - Reduces the number of bits to be transmitted
  - multimedia data transmission – such as text, audio and video

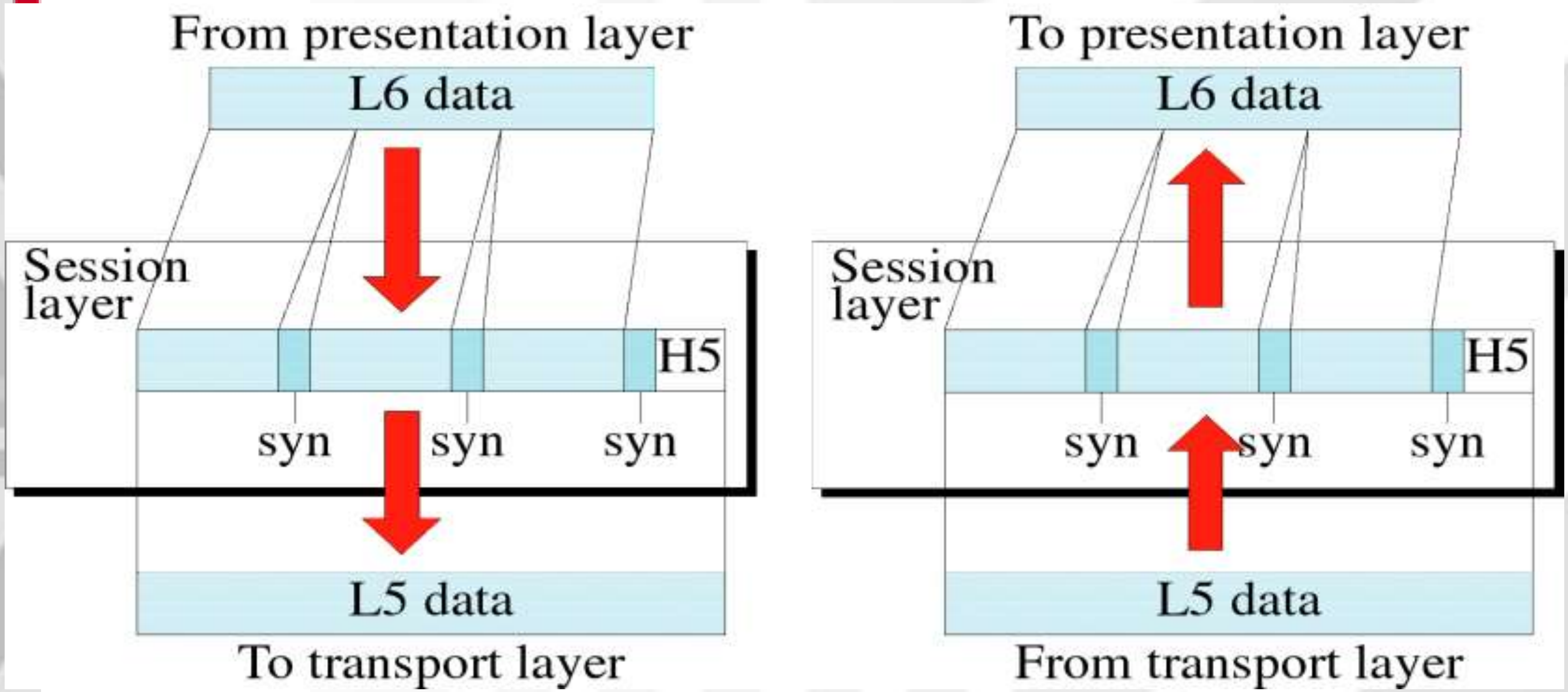
Presentation Layer



3.2 Session Layer

- **Session Layer (L5)**
  - The network *dialog controller*
  - *Establishes*, *maintains*, and *synchronizes* the interaction between communicating systems
- **Responsibilities:**
  - Dialog control
    - allows two systems to enter into a dialog
    - communication between two process – half-duplex or full-duplex
  - Synchronization
    - allows a process to add checkpoints (synchronization points) into a stream of data
    - E.g.: sending a file..

Session Layer



3.2 Transport Layer

- **Transport Layer (L4):**
  - Review of a network layer responsibility:
    - Responsible for source-to-destination (end-to-end) delivery of the entire message
    - Individual packet – treats each packet independently
  - transport layer
    - Ensures the whole (entire) message arrives intact and in order
    - Oversee both error control and flow control at source-to-destination level
  - To added security, transport layer create a <u>connection</u> between the two end ports
    - Connection - Single logical path between the source and destination

3.2 Transport Layer

- Creating connection involves 3 steps:
  - Connection establishment
  - Data transfer
  - Connection release
- Has more control over sequencing, flow, error correction and detection

3.2 Transport Layer

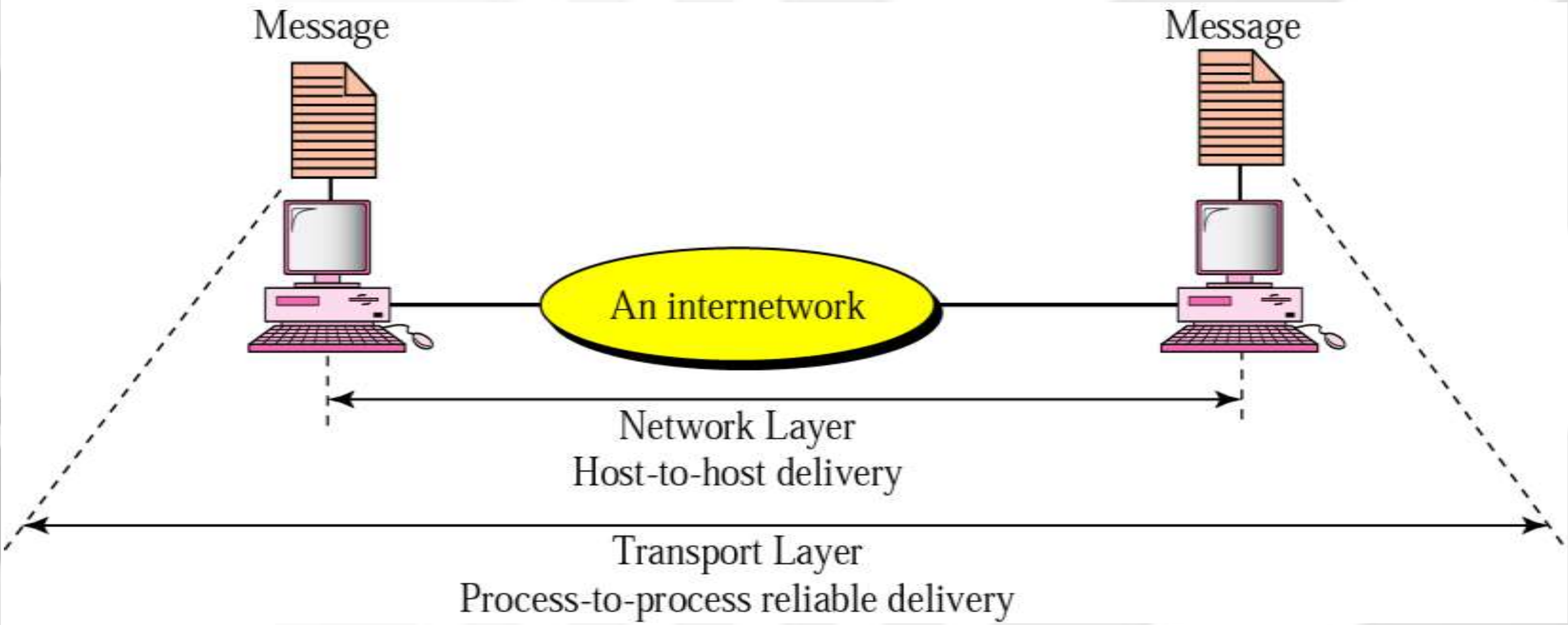
- Specific responsibilities:
  - Service-point addressing
    - Computers often run several programs at the same time
    - From a specific process (running program) on one computer to a specific process (running program) on the other
    - TL header must include a <u>service-point address</u> or <u>port address</u>
  - Segmentation and reassembly
    - Segment – add a sequence number into message segment
  - Connection control
    - Can be either connectionless (independent packet) or connection oriented
  - Flow control
    - End-to-end flow control (across multiple networks)
  - Error control
    - End-to-end error control (across multiple networks)

Note:

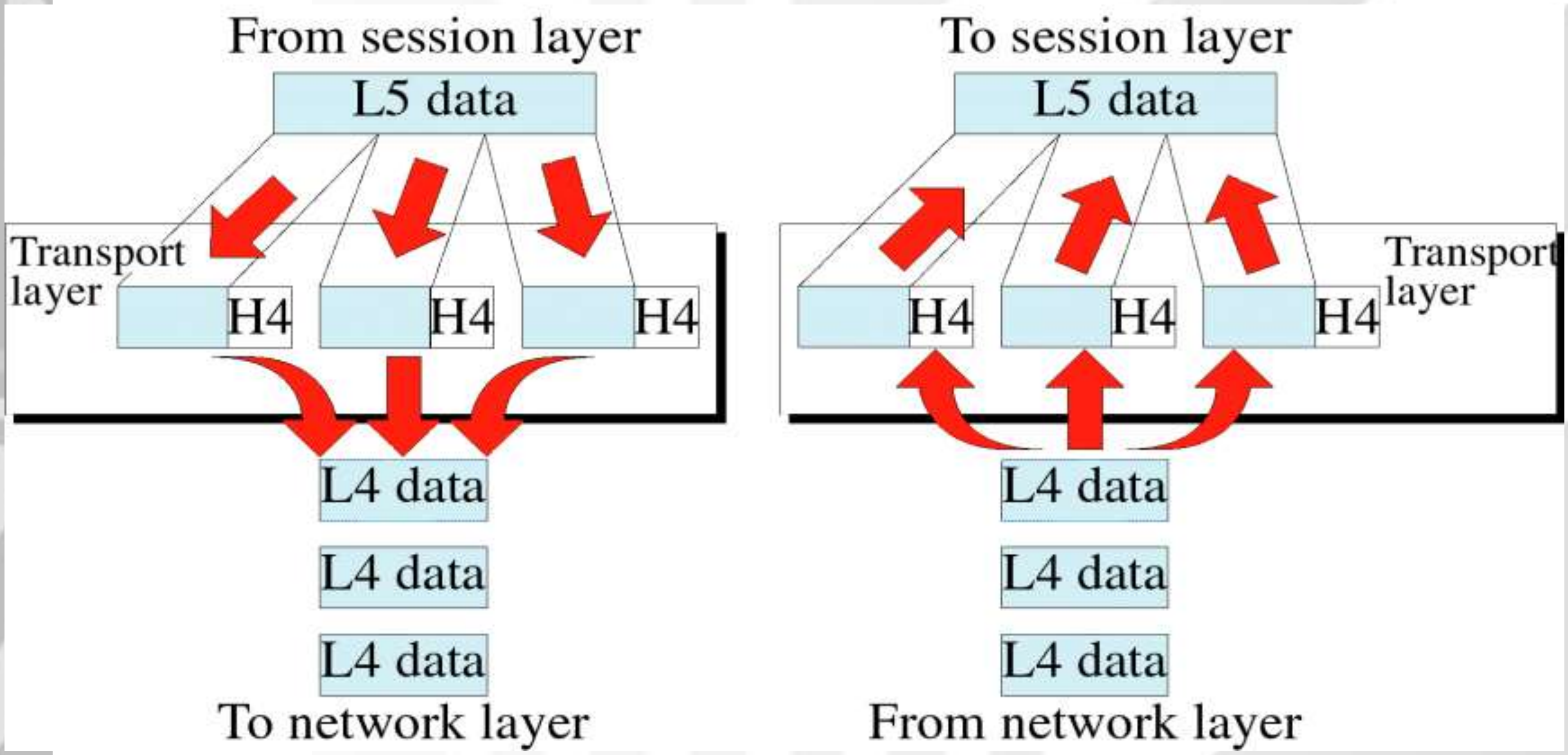
The transport layer is responsible for delivery of a message from one process to another.

Figure : Reliable process-to-process delivery of a message

http daemon守护进程



Transport Layer



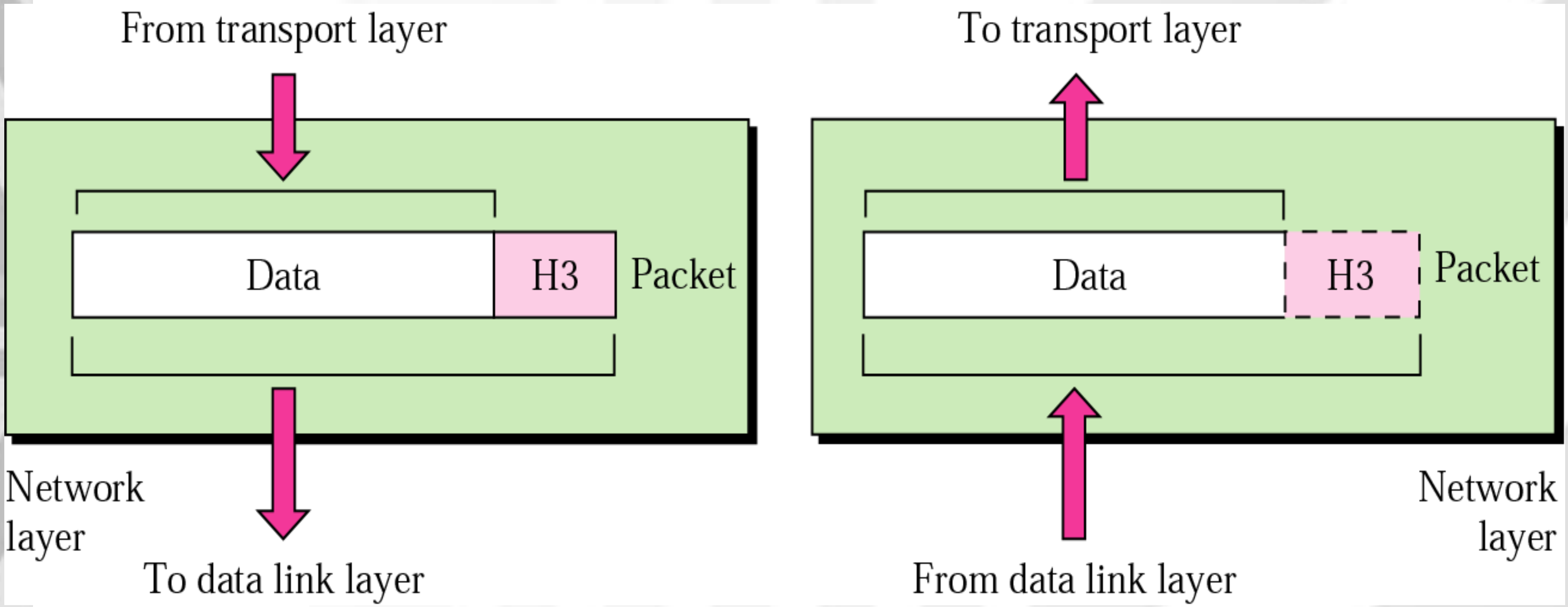
3.2 Function of Each Layer

- **Network Layer (L3):**
  - Responsible for the source-to-destination delivery of a packet possibly across multiple networks (links)
  - If two systems are attached to different networks, we need the network layer protocol to accomplish source-to-destination delivery
- **Specific responsibility:**
  - Logical addressing – to distinguish the source and destination systems when a packet passes the network boundary – also known network address
  - Routing – internetwork/large network – route the packet to the final destination

Note:

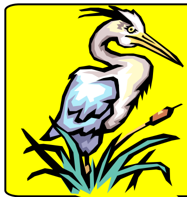
The network layer is responsible for the delivery of packets from the original source to the final destination.

Network Layer



Function of Each Layer

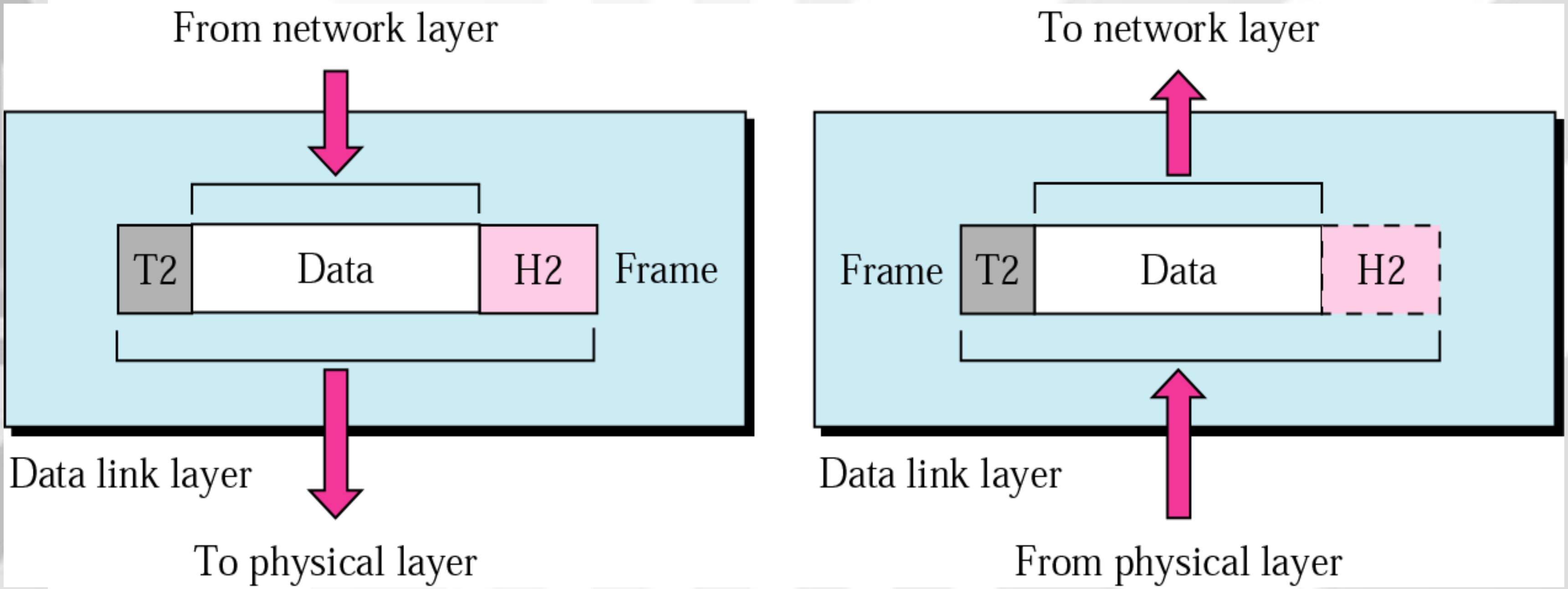
- **Data Link Layer (L2):**
  - Responsible for node-to-node delivery
  - Makes appear error free to the network layer
- **Responsibilities include:**
  - Framing – divides the stream data to manageable data units – frame
  - Physical addressing – adds a header to the frame –to define the physical address of sender(source address) and receiver (destination address)
  - Flow control – to prevent overwhelming at the receiver
  - Error control – provides reliability – to detect and retransmit damaged or lost frames, also prevent duplication of frames - trailer
  - Access control – require a protocol to determine which device has control over the link at any given time –> same link with two or more devices connected.



Note:

The data link layer is responsible for transmitting frames from one node to the next.

Data Link Layer

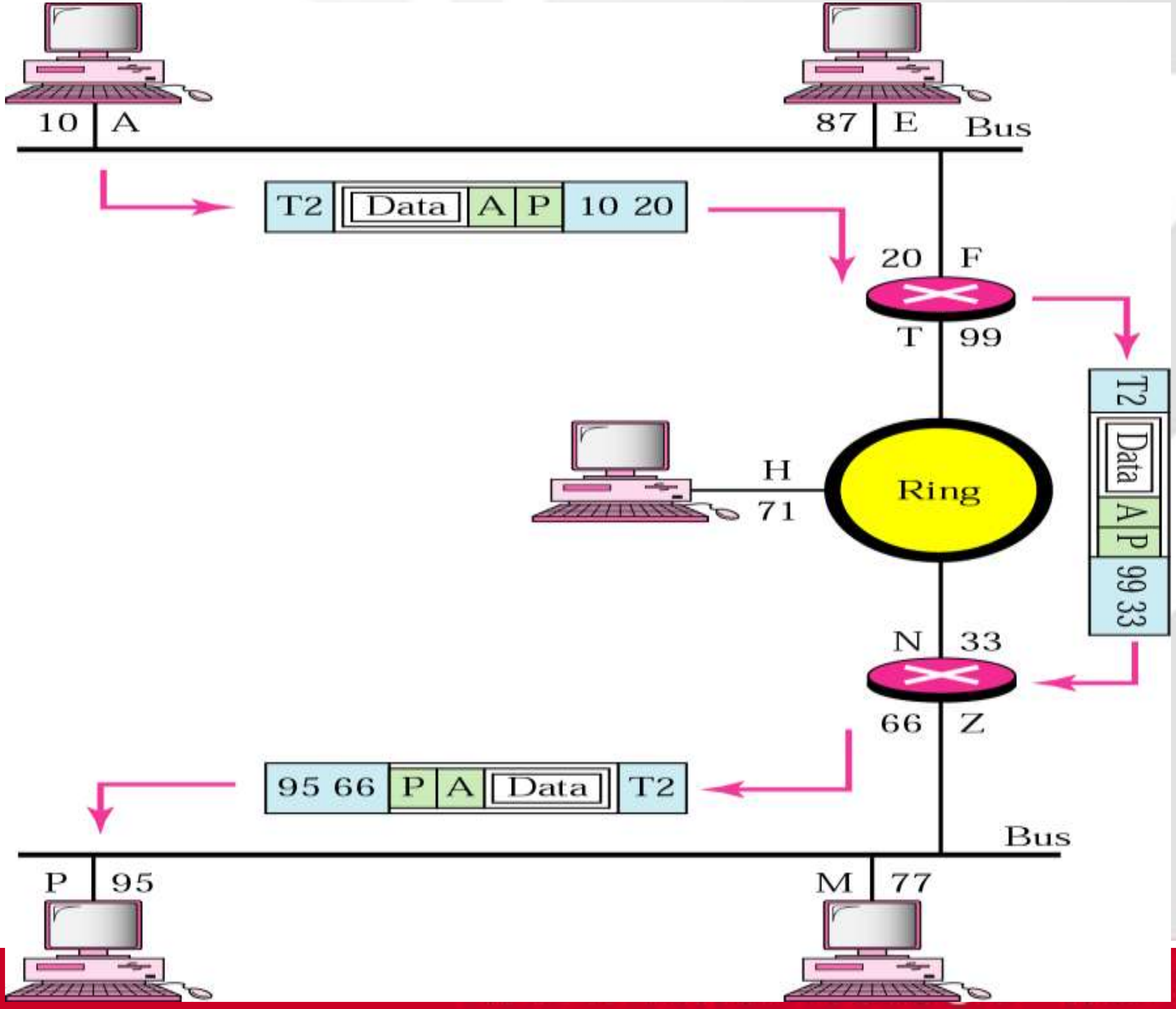


Network Layer Example

Example 1- if Dest is on different network

In the next figure, we want to send data from a node with network address A and physical address 10, located on one LAN, to a node with a network address P and physical address 95, located on another LAN. Because the two devices are located on different networks, we cannot use physical addresses only; the physical addresses only have local jurisdiction. What we need here are universal addresses that can pass through the LAN boundaries. The network (logical) addresses have this characteristic.

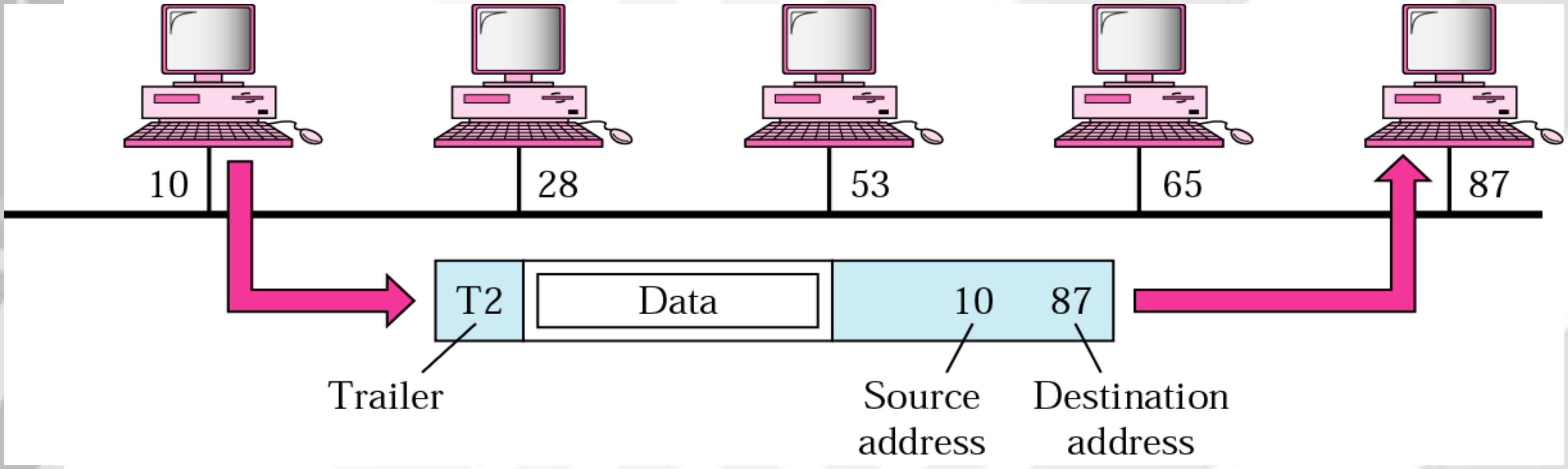
Network Layer Example



Example 1- if dest in same network

In the next figure, a node with physical address 10 sends a frame to a node with physical address 87. The two nodes are connected by a link. At the data link level this frame contains physical addresses in the header. These are the only addresses needed. The rest of the header contains other information needed at this level. The trailer usually contains extra bits needed for error detection

Data Link Layer Example

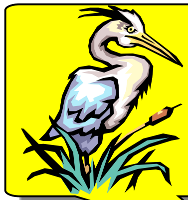


Function of Each Layer

- Physical layer (L1)
  - Coordinates the function required to transmit a bit stream over a physical medium
  - Deal with the mechanical/electrical spec of the interface and transmission medium
  - Defines the procedures and functions that physical devices and interfaces have to perform for transmission to occur
- Concerned with:
  - Physical characteristics of interfaces and media
  - Representation of bits
    - Bits must be encoded into signals –electrical or optic
  - Data rate –the number of bits sent each second
  - Synchronization of bits
  - Line configuration – connection of devices to the medium

Function of Each Layer

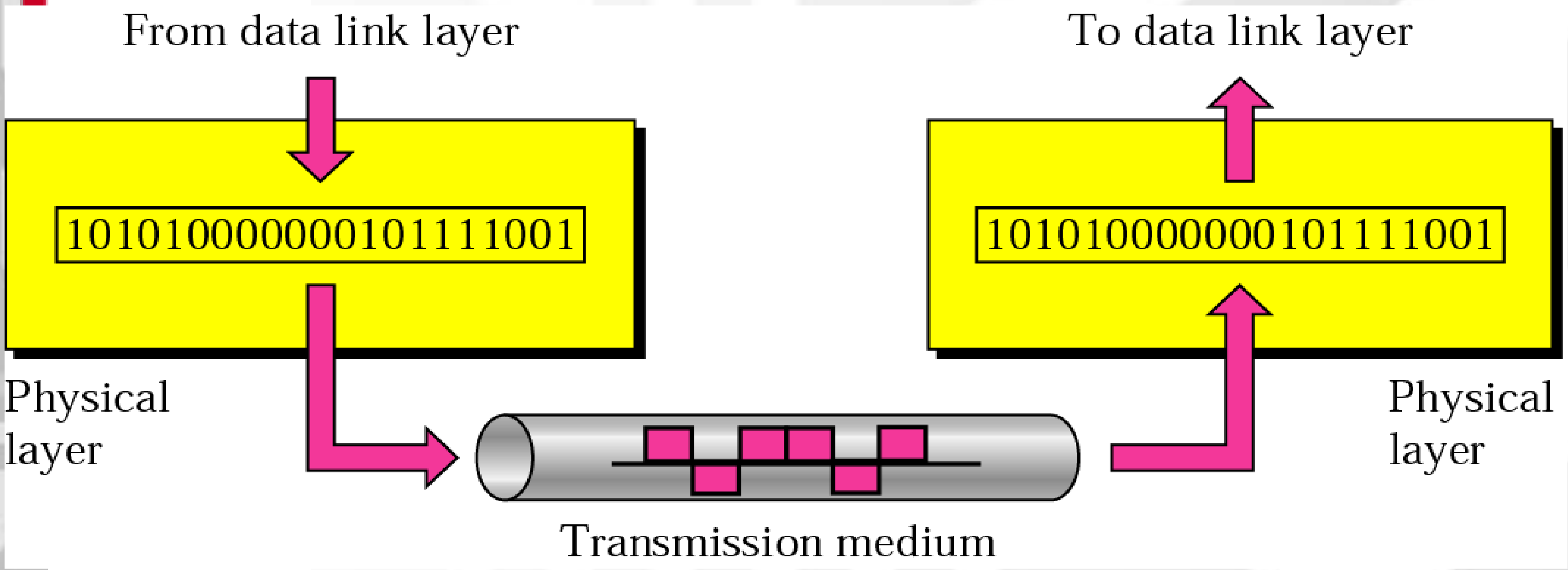
- Concerned with:
  - Physical topology
    - How devices are connected to form a network
  - Transmission mode
    - Direction of signal transmission between two devices



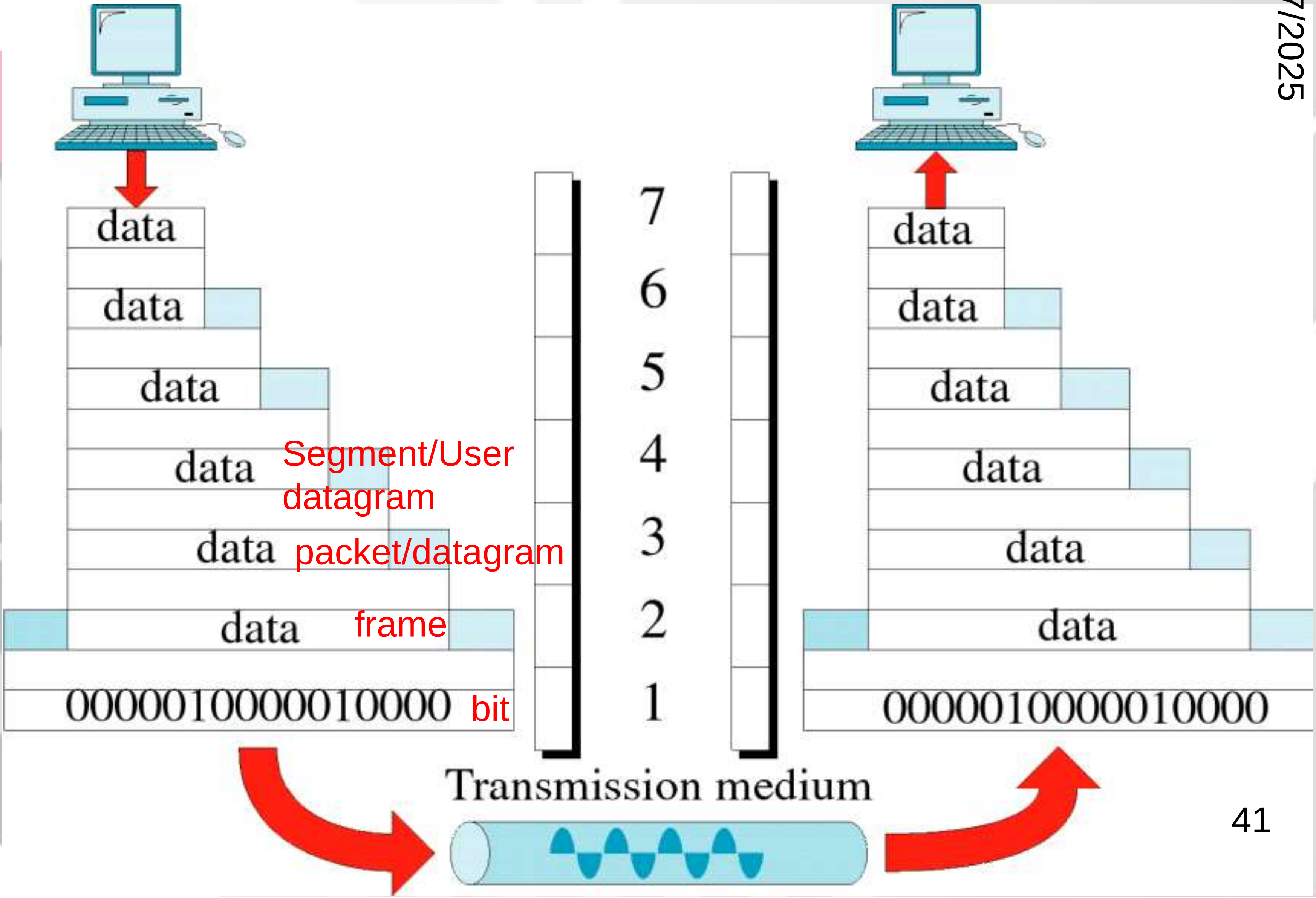
Note:

The physical layer is responsible for transmitting individual bits from one node to the next.

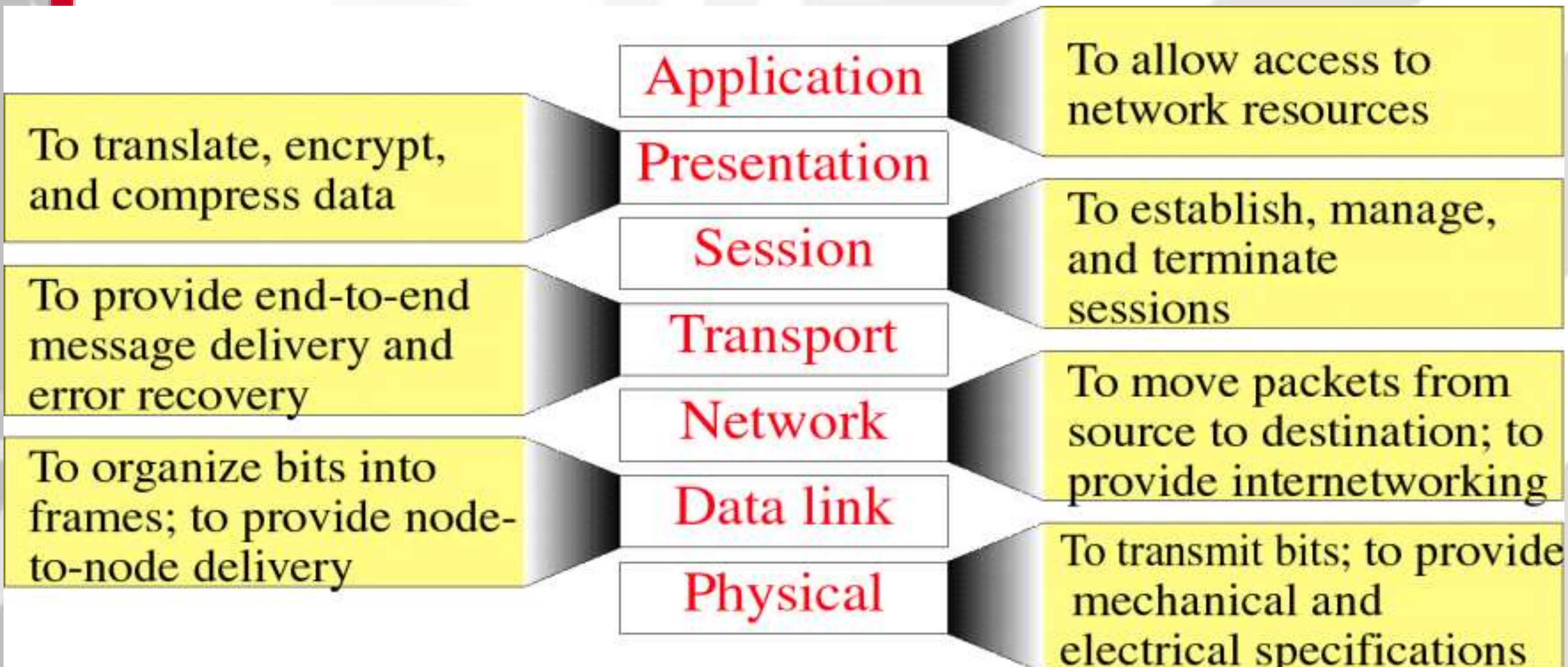
Physical Layer



An Exchange Using the OSI Model



Summary of Layer Functions (OSI model)



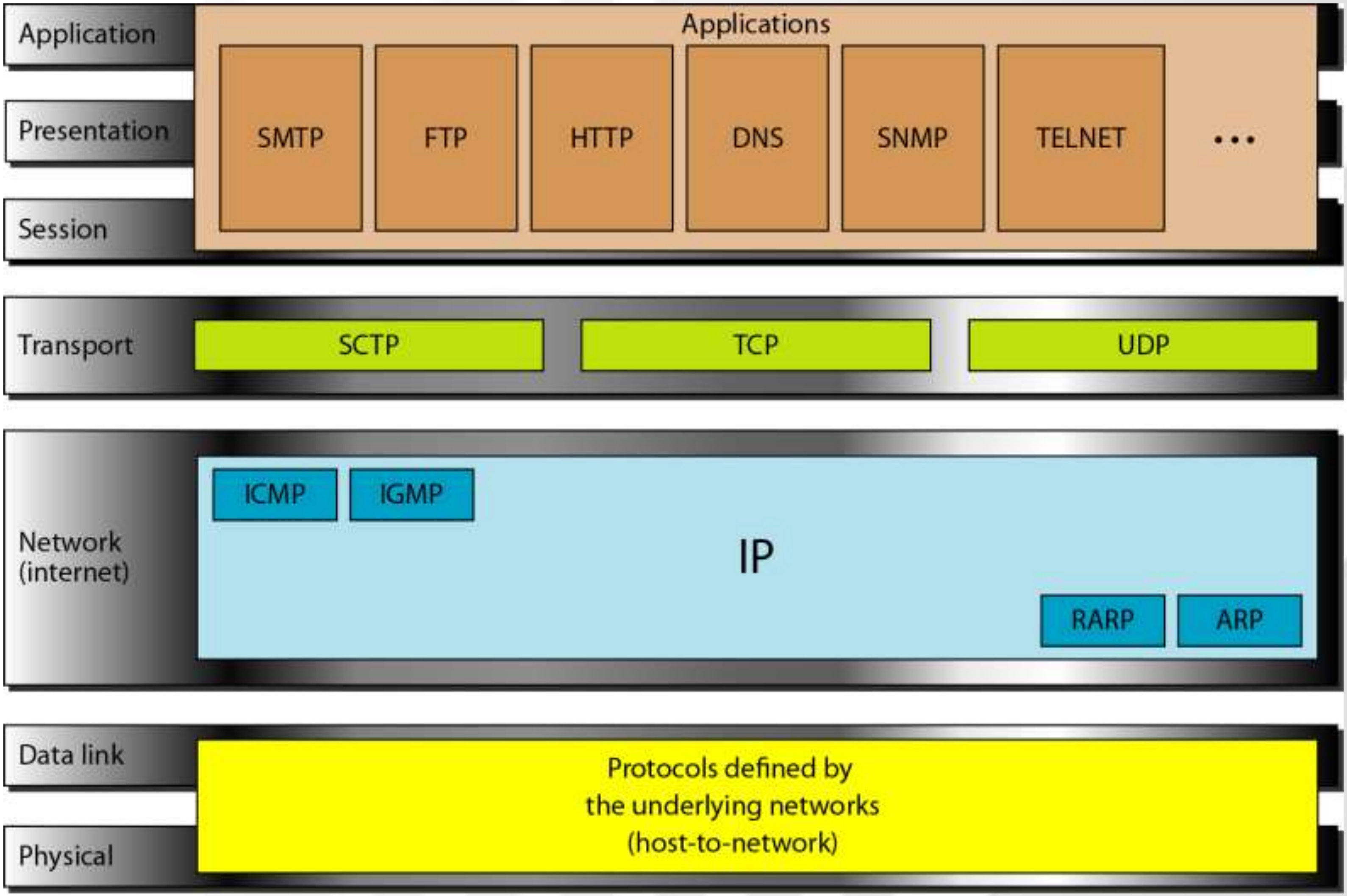
3.3 TCP/IP Protocol Suite

- Developed prior to the OSI model
- 5 layers – also known Internet model
- The three topmost layers in the OSI model are represented in TCP/IP by a single layer – application layer
- TCP/IP is a hierarchical protocol – the upper-level protocol is supported by one or more lower-level protocols
- E.g.: @ TL – TCP, UDP; @NL - IP

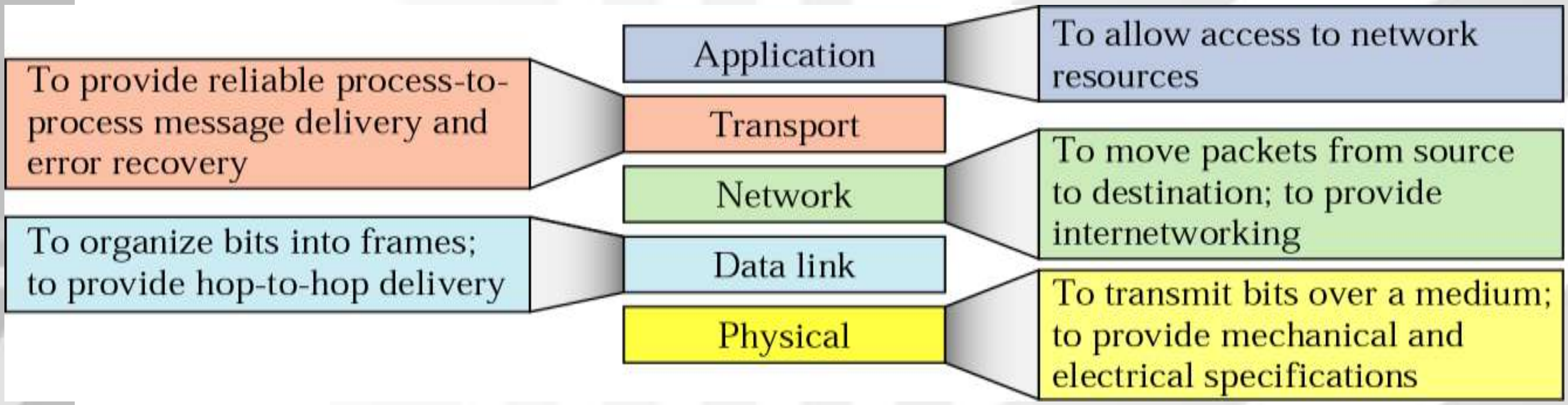
TCP/IP

- Physical layer
  - Twisted pair, optical fibers, satellite
- Data link layer
  - Ethernet, WiFi
- Network layer
  - IP
- Transport layer
  - TCP, UDP, SCTP
- Application Layer
  - SMTP, FTP

Figure 2.16 *TCP/IP and OSI model*

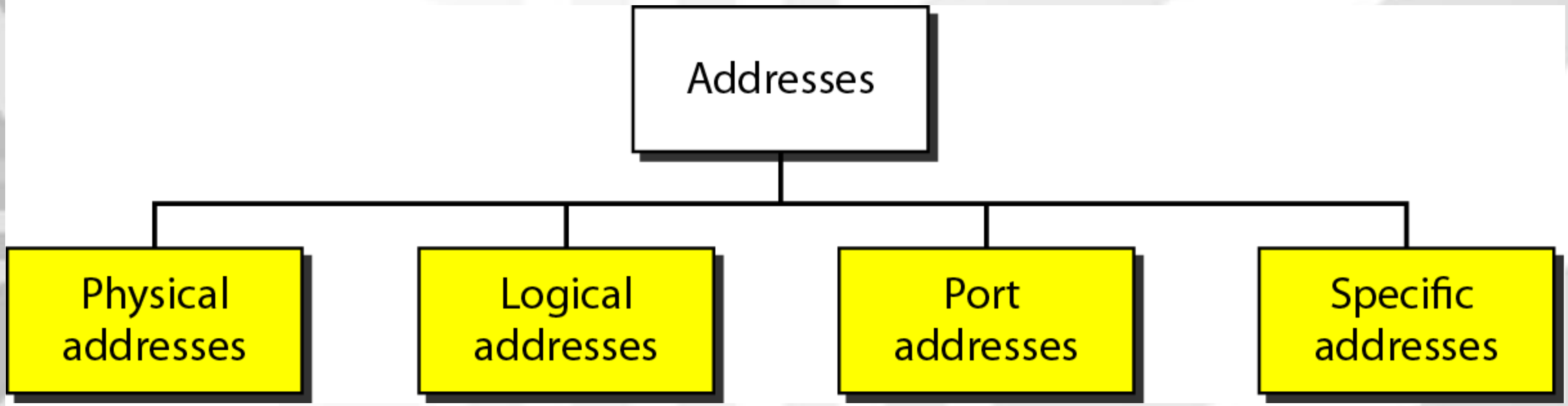


Summary of Layer Functions (TCP/IP)



Example of using TCP/IP

Prior to the example, you need to know the following terminologies in which will mapped to the TCP/IP model



Addresses in TCP/IP

Relationship of layers and addresses in TCP/IP